The Thermodynamic Cost of Knowing

Observation as Irreversible Payment — A Formally Verified Bridge from Quantum Measurement to Classical Thermodynamics

$$V^2 + \big(1 - H_{\text{bits}}(\rho)\big) \le 1 \qquad \Longrightarrow \qquad Q \ge k_B T \ln 2 \cdot \big(1 - V^2\big)$$

Every fraction of a bit of which-path information causes a proportional destruction of interference—and each fraction carries an exact Landauer cost.

TECHNICAL NOTE —— March 2026

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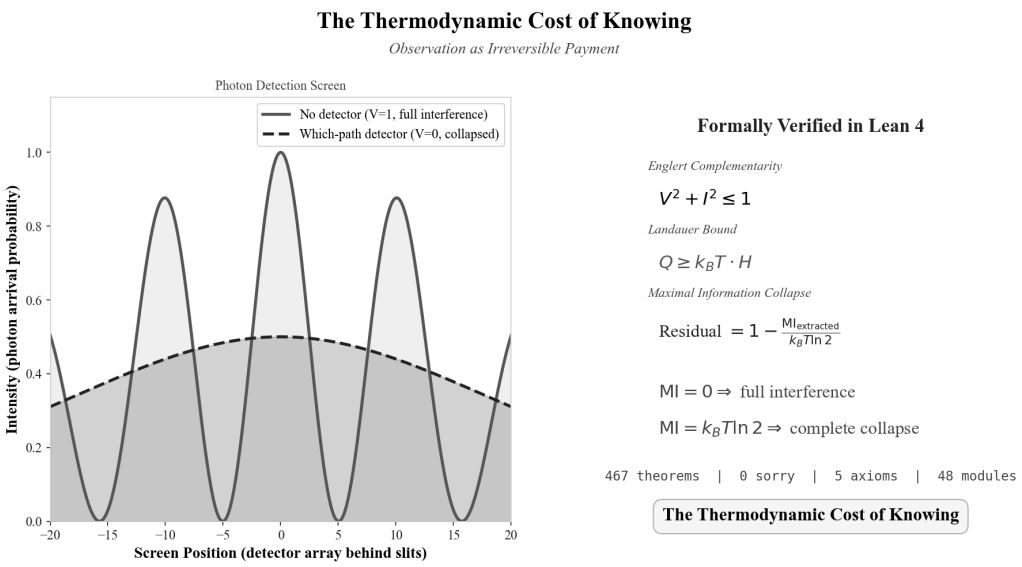
Lean 4 + Mathlib, Haskell QuickCheck, Coq/Rocq, Agda, Python



Core Result: Principle of Maximal Information Collapse. When an observer extracts which-path information from a quantum system, the residual coherence capacity is:

$$\text{Residual Coherence} = 1 - \frac{\text{MI}_{\text{extracted}}}{k_B T \ln 2} \in [0, 1]$$

Extract 0 bits $\Longrightarrow$ full interference. Extract 1 bit $\Longrightarrow$ complete decoherence.

Crucially, observation is not binary. Partial measurement extracts a *fraction* of a bit, causing proportional decoherence along the Englert curve $V = \sqrt{1 - I^2}$.

Machine-checked in Lean 4 with Mathlib. ***467 theorems, 0 `sorry`, 5 explicit axioms*** *(corresponding to physical principles not yet formalized in Mathlib).*

1 Motivation and Prior Work

The double-slit experiment is the canonical demonstration of quantum mechanics [5, 6]: a single particle interferes with itself, producing fringes on a detection screen—unless an observer determines *which slit* the particle traversed, in which case the fringes vanish [4]. This complementarity between path information and interference visibility was first quantified by Wootters

and Zurek [7] and Greenberger and Yasin [8], then sharpened to Englert's tight inequality $V^2 + I^2 \leq 1$ [9].

Though well established, this tradeoff is typically treated as a purely quantum phenomenon, disconnected from classical thermodynamics. The decoherence programme [13, 14] explains the *mechanism* of the quantum-to-classical transition but does not quantify its *thermodynamic cost*. This gap leaves a fundamental question open: what is the minimum energy required, in joules, to destroy a given fraction of interference?

Our prior work on the **Unified Material-State Tensor (UMST)** framework [1, 2, 3] established a formally verified thermodynamic gate for continuum materials: a Kleisli-categorical structure enforcing conservation laws (mass, energy, hydration bounds) with graded admissibility. The key insight was that *admissibility is compositional but not transitive*—two individually valid transitions can compose to an inadmissible one, proved via triangle-inequality grading (the `admissibleTrans` counterexample: densities $\{0, 99, 198\}$ kg/m^3 with $\delta = 100$).

The present work extends this framework to quantum measurement. We show that the act of measurement is not a mysterious "collapse" but a *thermodynamic transaction*: extracting which-path information requires physical work bounded below by Landauer's principle [10, 11] ($Q \geq k_B T \ln 2$ per bit, experimentally confirmed [12]), and this energy expenditure is *mathematically identical* to the destruction of quantum coherence.

2 Mathematical Foundations

2.1 The Quantum State (Path Qubit)

The formalization models the spatial degree of freedom of a particle traversing a double-slit interferometer as a two-level quantum system (a qubit). The state is represented by a **density matrix** $\rho \in \mathbb{C}^{2\times 2}$ satisfying three constraints:

1. **Hermitian:** $\rho = \rho^\dagger$ (the matrix equals its conjugate transpose).
2. **Positive semi-definite (PSD):** For any state vector $|\psi\rangle$, $\langle\psi|\rho|\psi\rangle \geq 0$.
3. **Unit trace:** $\mathrm{Tr}(\rho) = \rho_{00} + \rho_{11} = 1$.

The diagonal elements are the classical Born weights—probabilities of traversing each slit:

$$p_0 = \rho_{00}, \qquad p_1 = \rho_{11}, \qquad p_0, p_1 \geq 0, \qquad p_0 + p_1 = 1$$

The off-diagonal element $\rho_{01} = \rho_{10}^*$ is the complex **coherence**: the quantum interference capability. Its magnitude $|\rho_{01}|$ directly controls fringe contrast; its phase determines the spatial alignment of the interference pattern.

2.2 Path Distinguishability and Fringe Visibility

Two physical observables characterize the double-slit complementarity:

Definition 1 (Path Distinguishability)**.** $I = |p_0 - p_1| \in [0, 1]$. *When $p_0 = p_1 = \frac{1}{2}$, the observer has zero which-path knowledge ($I = 0$); when $p_0 = 1$ and $p_1 = 0$, the path is fully determined ($I = 1$).*

Definition 2 (Fringe Visibility)**.** $V = 2|\rho_{01}| \in [0, 1]$. *This is proportional to the magnitude of quantum coherence. $V = 1$ corresponds to maximal interference fringes; $V = 0$ means no interference pattern at all.*

2.3 Englert's Complementarity Relation

The inequality $V^2 + I^2 \leq 1$ is not a postulate—it is a *theorem* derivable from the PSD constraint alone. Because ρ is positive semi-definite, its determinant is non-negative:

$$\det(\rho) = \rho_{00}\rho_{11} - |\rho_{01}|^2 \geq 0 \qquad \Longrightarrow \qquad |\rho_{01}|^2 \leq p_0 p_1$$

Substituting into the definitions of V and I:

$$V^2 + I^2 = 4|\rho_{01}|^2 + (p_0 - p_1)^2 \leq 4p_0p_1 + (p_0 - p_1)^2 = (p_0 + p_1)^2 = 1$$

This is the content of the Lean theorem `complementarity_fringe_path` in module `QuantumClassicalBridge`. Any gain in which-path distinguishability I mathematically necessitates a rigid destruction of interference visibility V.

3 The Measurement Channel

The act of a detector at the slits is modeled as a **Kraus trace-preserving completely positive map**. An ideal which-path interaction is the Lüders dephasing channel:

$$\mathcal{E}_{\text{path}}(\rho) = \Pi_0 \rho \Pi_0 + \Pi_1 \rho \Pi_1$$

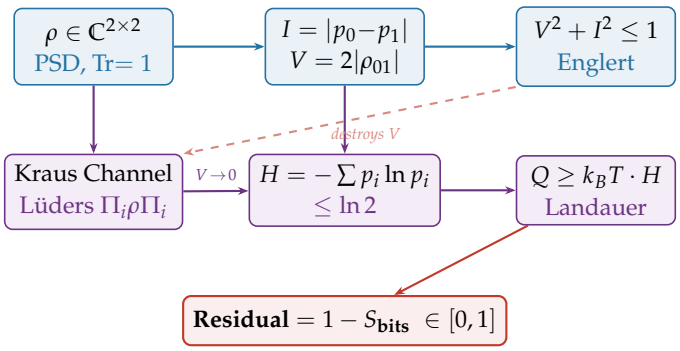
where $\Pi_0 = |0\rangle\langle 0|$ and $\Pi_1 = |1\rangle\langle 1|$ are the spatial projectors onto each slit. These satisfy the Kraus properties proved in the repository (`MeasurementChannel.lean`):

- **Self-adjoint:** $\Pi_i = \Pi_i^\dagger$
- **Idempotent:** $\Pi_i^2 = \Pi_i$
- **Orthogonal:** $\Pi_0 \Pi_1 = 0$
- **Complete:** $\Pi_0 + \Pi_1 = \mathbb{I}$ (trace-preserving)

This channel preserves the diagonal entries (p_0 and p_1 are unchanged) but strictly forces the off-diagonal coherences to zero ($\rho_{01} \to 0$), which forces $V \to 0$ post-measurement. This is the formal content of wavefunction collapse on the path basis: not a mysterious event, but a mathematically precise trace-preserving operation whose thermodynamic cost we quantify next.

4 Proof Architecture

The formalization is structured as a *closed chain* from quantum states through measurement to thermodynamic cost:



Top row (cyan): The quantum state ρ yields observables I and V, bounded by Englert complementarity. **Middle row (purple):** The Lüders channel extracts which-path information, producing diagonal entropy H and incurring a Landauer cost. **Bottom (red):** The Principle of Maximal Information Collapse unifies these into a single residual-coherence metric.

5 The Principle of Maximal Information Collapse

5.1 Diagonal Information Entropy

The diagonal von Neumann entropy quantifies the classical uncertainty in the Born weights:

$$H = -p_0 \ln p_0 - p_1 \ln p_1 \qquad (\leq \ln 2 \text{ nats})$$

This is the Shannon entropy of the computational-basis probabilities, capped at $\ln 2$ for a binary system. We normalize this to bit-equivalents:

$$S_{\text{bits}} = \frac{H}{\ln 2} \in [0, 1]$$

5.2 The Landauer Cost

Extracting the which-path bit from the quantum system to a macroscopic register incurs a strict physical lower bound on heat dissipation via Landauer's Principle [10]:

$$Q \geq k_B T \ln 2 \cdot S_{\text{bits}}$$

This cost is bounded above by a single Landauer bit-energy $k_B T \ln 2$, because the path qubit carries at most one bit of information. The bound has been experimentally confirmed to within statistical error [12].

5.3 Residual Coherence Capacity

The **residual coherence capacity** after information extraction is formally proved to satisfy:

$$\text{Residual} = 1 - S_{\text{bits}} \in [0, 1]$$

with tight bounds at both endpoints:

- $S_{\text{bits}} = 0$ (no information extracted) $\implies$ Residual $= 1$ (full visibility possible).
- $S_{\text{bits}} = 1$ (maximum 1-bit extraction) $\implies$ Residual $= 0$ (complete decoherence).

We also prove the **cost–coherence identity**:

$$Q = k_B T \ln 2 \cdot (1 - \text{Residual})$$

The thermodynamic cost is *exactly proportional* to the fraction of coherence destroyed. A concrete `ErasureProcess` structure in Lean enforces dissipatedHeat $\geq$ landauerCostDiagonal via the Second Law.

5.4 The Continuous Tradeoff

The standard textbook narrative presents measurement as all-or-nothing: "observe and the fringes vanish." The Englert relation reveals a *continuum*: a probe that extracts $I = 0.3$ bits of which-path information reduces fringe visibility to $V = \sqrt{1 - 0.09} \approx 0.95$—the interference pattern is barely disturbed. At $I = 0.7$ the fringes are heavily suppressed ($V \approx 0.71$). Full decoherence requires extracting the *entire* bit. Every point on the curve $V^2 + I^2 = 1$ is physically realizable, and each carries a proportional Landauer cost. This continuous tradeoff is precisely what the residual-capacity framework captures and what the Lean formalization proves.

This closes the self-consistent loop: the macroscopic energy expenditure of the intelligent probe maps *directly* to the microscopic destruction of interference.

6 What Is Formally Proved

Every row below is a machine-checked theorem in Lean 4 with Mathlib [16, 17]—no `sorry` in these modules. The repository contains one Phase-5 placeholder: general unital DPI / Klein in `DataProcessingInequality.lean` (see `Lean/VERIFY.md`).

Theorem	Statement	Module
Englert complementarity	$V^2 + I^2 \leq 1$	`QuantumClassicalBridge`
Which-path collapse	$V \to 0$ after Lüders channel	`MeasurementChannel`
Projector properties	self-adjoint, idempotent, orthogonal, TP	`MeasurementChannel`
Density matrix diagonals	PSD $\implies p_i \geq 0, \sum p_i = 1, p_i \leq 1$	`DensityState`
Diagonal entropy bound	$H_{\text{diag}} \leq \ln 2$	`InfoEntropy`
Landauer cost cap	cost $\leq k_B T \ln 2$	`LandauerBound`
Path entropy ≤ 1 bit	$S_{\text{bits}} \in [0, 1]$	`LandauerBound`
Maximal collapse	$S_{\text{bits}} = 1 \implies \text{Residual} = 0$	`LandauerBound`
Null preservation	$S_{\text{bits}} = 0 \implies \text{Residual} = 1$	`LandauerBound`
Cost–coherence identity	$Q = k_B T \ln 2 \cdot (1 - \text{Residual})$	`LandauerBound`
Erasure $\geq$ bound	dissipatedHeat $\geq$ landauerCostDiagonal	`LandauerBound`
Which-path invariance	Landauer cost unchanged by measurement	`LandauerBound`
Gate enforcement	admissibility + Landauer + cap in one	`DoubleSlit`

7 Assumptions and Scope

The formalization makes the following explicit assumptions—none of which are proof gaps in the stated formal theorems. They are *scope control*: boundaries on what the current verification covers.

1. **Standard finite-dimensional QM** as encoded in Mathlib: complex matrices, `PosSemidef`, trace, Kraus form for CPTP maps.
2. **Path qubit only:** Hilbert space `Fin 2` $\to \mathbb{C}$ for the which-path degree of freedom. No spatial fringe pattern in position/momentum bases within this formal layer.
3. **Coarse readout** (I, V)**:** Defined proxies, not derived von Neumann entropy of an apparatus.
4. **Diagonal entropy:** `vonNeumannDiagonal` uses computational-basis diagonals; for general pure states with coherence, this can differ from the full $S(\rho) = -\text{Tr}(\rho \log \rho)$.
5. **Landauer layer:** `landauerCostDiagonal` is a scale $k_B T \ln 2 \times$ (diagonal entropy in bit-equivalents)—not a claim that measured heat equals this without an explicit erasure/bath model.

What is not claimed:

- Derivation from a relativistic or gravitational UMST layer.
- Identification of I with a specific experimental which-path meter without calibration axioms.
- Quantum DPI / mutual information on compound systems (listed as future in `InfoEntropy`).
- Mixed ensembles as formal convex sums in `DensityState` (future).
- Certified ODE/PPO solver convergence/stability claims without a dedicated numerical analysis layer.

8 Connection to the UMST Framework

This work extends the UMST formal verification stack in three directions:

1. **From classical to quantum admissibility.** The original UMST gate checks mass conservation, Clausius-Duhem entropy, and hydration bounds on *classical* states (over $\mathbb{Q}$ with `AdmissibleN`). Here, the gate operates on *quantum* density matrices: the projector $\Pi_i \rho \Pi_i$ is the measurement-channel analogue, and admissibility is enforced by Englert complementarity plus the Landauer bound.

2. **From material hydration to information erasure.** In the original framework, the Helmholtz free energy $\psi(\alpha)$ governs the cost of hydration transitions. Here, the diagonal von Neumann entropy $H(\rho)$ governs the cost of information extraction. Both are monotone hooks: increasing hydration lowers free energy; increasing path information lowers residual coherence. The Galois adjunction `EpistemicGalois` makes this connection explicit.

3. **From counterexample to principle.** The discovery that `admissibleTrans` is *refutable* led to graded admissibility. The analogous quantum insight is that partial measurements compose non-trivially: two weak measurements each extracting < 1 bit can together extract the full bit, collapsing coherence entirely. The residual-capacity framework captures this compositional structure.

9 Verification Stack

Layer	Tool	Scope	Status
Formal proof	Lean 4 + Mathlib	48 modules, 467 theorems	**0 sorry, 5 axioms**
Property testing	Haskell QuickCheck	14 properties + sanity	**All pass**
Numerical sim	Python (stdlib + QuTiP)	62 unit tests, 4 sims	**All pass**
Cross-language	Coq + Agda	LandauerEinsteinBridge	**0 Admitted**

The Python simulator demonstrates the collapse visually: as which-path information I increases from 0 to 1, the interference pattern progressively flattens, tracking the $V = \sqrt{1 - I^2}$ Englert curve exactly. The accompanying animated GIF (`Docs/Media/double-slit-collapse.gif`) visualizes this continuous tradeoff across 60 frames.

Repository: `github.com/tytolabs/umst-formal-double-slit` | *License:* MIT | *Build:* `cd Lean && lake build`

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Acknowledgments. Portions of this work were developed in collaboration with advanced large-language-model tools. Claude Opus and Sonnet 4.6 (Anthropic) provided surgical precision during drafting and refinement. Gemini 3.1 Pro High (Google) offered exceptional large-context planning and file management. Grok 4.20 (xAI) and its collaborative reasoning team contributed core mathematical and scientific reasoning. The Cursor code editor, Claude Code (Anthropic), and Antigravity (Google) supported seamless implementation.

The large-language models assisted with exploration, drafting, and code scaffolding—never with the validity of formal proofs. All theorems were machine-checked by their respective compilers (Lean 4, Coq/Rocq, Agda), which accept only well-typed terms, never persuasive arguments.

This work relies on the following open-source tools and languages: Lean 4 (theorem prover), Mathlib (mathematical library), Coq/Rocq (proof assistant), Agda (dependently typed language), Haskell (property testing via QuickCheck), Python (numerical simulation), QuTiP (quantum toolkit), LaTeX (typesetting), Git/GitHub (version control and CI), and GitHub Actions (continuous integration).

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